RELEASE HISTORY

Power Glove Vision Changelog

Versioned features, fixes, security changes, and documentation updates.

2026 EDITION / IAIN BENNETT /

This file records user-visible PowerGlove Vision changes. The project follows [Keep a Changelog] categories and uses [Semantic Versioning]. Git remains the authoritative record for line-level and file-level history.

Unreleased

Added

- Added a live active-profile selector to the Dashboard without changing the startup profile saved on Setup.
- Added a dedicated matrix gestures-idle state with a pinball-style animated glove, separate from both true shutdown and the flashing error X.

Changed

- Made **Gestures off** a healthy worker state that releases controller input, closes the camera and MediaPipe tracker, and keeps the website and authenticated RetroPie profile listener available.
- Made camera and model initialization lazy so idle mode performs no capture or vision processing and can return to an active profile without restarting the website.
- Reworked the matrix attract sequence into distinct pinball-style beats: a four-frame energy sweep, a broad travelling cuff, a staged glove reveal, intermediate finger curls, an eight-position spark with a comet trail, one outline pulse, and a readable hold.
- Used the UNO Q matrix's full eight-level grayscale range to separate the dim glove body, spark halo, bright spark, and whole-glove pulse.

Documentation

- Documented the UNO Q matrix as an eight-level monochrome DMD/BitPixel-style design target, including silhouette, contrast, motion, pulse, and physical review guidance for future animations.
- Standardized source headers with each file's purpose, author, copyright, SPDX license identifier, local history, and links to the complete history.
- Documented public interfaces and non-obvious security, lifecycle, tracking, packaging, and rendering functions.
- Added this centralized project changelog and its print-ready PDF edition.
- Added an automated audit for required source headers, module descriptions, and production Python interface docstrings.
- Added a complete configuration reference covering JSON templates and active copies, gesture thresholds, manifests, RetroArch mapping, systemd units, generated state, secret handling, and the App Lab installation ZIP location.
- Added GitHub Actions checks for supported Python versions, tests, source and documentation audits, syntax, PDF builds, and App Lab installation ZIP verification.
- Added a security policy for private reporting, pairing and network boundaries, shared-token handling, shutdown permissions, and dependency integrity.
- Added a contributing guide for code style, tests, documentation, changelog, generated artifacts, commits, and pull requests.
- Added reusable documentation and App Lab installation package verification tools.
- Added an illustrated, one-page-per-game handbook for all eight automatically configured titles, with original direction, finger-pose, wrist, and depth art.

- Placed cropped gesture illustrations beside the corresponding instructions in the gameplay handbook, including the universal V-sign and thumbs-up controls.
- Added the same contextual gesture illustrations to every Program A-I card and documented the menu gestures shared by all nine programs.
- Added an off-script gameplay section that encourages safe experiments with other NES and Famicom games, especially the unassigned A, D, and H programs.
- Updated the cabinet cheat sheet with off-script profile testing, safe behavior for unknown games, and exact ROM registration guidance.
- Renamed the built-in installation Help route, retained the original URL as an alias, and aligned its summaries with the Play Checklist terminology.
- Extended the documentation audit to require Help-library coverage and a gameplay section for every title in the shipped game registry.
- Expanded Wi-Fi deployment verification to confirm the current installation and gameplay guides, raw Markdown, and gesture artwork on the UNO Q.
- Fixed the Help renderer so the allowlisted gesture images embedded in control tables appear alongside their actions, while unsafe raw HTML remains escaped.
- Extended UNO Q deployment checks across every Help guide and representative gameplay and Programs A-I illustrations.
- Documented the required Markdown-to-PDF workflow and UNO Q Help synchronization steps for documentation contributions.
- Added offline PDF links to every public Help guide and the project overview, while keeping the cabinet-specific quick-reference PDF private.
- Included the nine allowlisted public PDFs in UNO Q deployments and App Lab installation packages, with package and live-route verification.
- Established `dev` as the integration branch, documented release and hotfix promotion into `main`, and enabled CI validation for pushes to both branches.
- Refreshed the Dashboard, Learn, and Setup screenshots from the running UNO Q after the tagline, compact diagnostics, and safe-shutdown controls were added.
- Added an offline Help library that renders the maintained public Markdown guides with responsive navigation, contents links, illustrations, tables, code samples, and access to the original source.
- Added a dynamic **This cabinet** Help page whose UNO Q links follow the browser's validated hostname or IP and whose non-secret RetroPie values come from the active configuration.
- Rewrote the configuration reference for public installations, with guided UNO Q and RetroPie setup, field-level behavior, safe gesture tuning, network boundaries, backup and recovery advice, and symptom-based troubleshooting.
- Renamed the Field Guide as the Installation Guide and generalized camera instructions for UVC-compatible USB cameras while recording the Razer Kiyo as tested reference hardware.

Fixed

- Changed Wi-Fi deployment to use SFTP staging and terminal-backed remote commands for UNO Q systems that stall non-terminal SSH sessions.

- Allowed the UNO Q deployment health check to use the board's current IP when its `.local` name pauses during a container restart.
- Published the host shutdown request atomically so a filesystem observer cannot consume the request between file creation and the final content write.
- Updated the GitHub Actions workflow to use the current Node 24 action releases and run the Python 3.7 compatibility job on Ubuntu 22.04.
- Corrected Program I so index curl accelerates in Knight Rider, a forward push accelerates with turbo, and thumb curl fires the weapons.

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Added

- Camera-only Power Glove tracking on Arduino UNO Q with MediaPipe.
- Gesture profiles for Bad Street Brawler, Super Glove Ball, and cartridge-free Programs A-I.
- Authenticated UDP profile selection and virtual Linux gamepad output for RetroPie, including per-game runcommand hooks.
- Dashboard, live diagnostics, offline gesture lessons, configuration controls, camera recovery, controller start/stop controls, and UNO Q matrix feedback.
- Wi-Fi deployment, App Lab packaging, runtime-asset retrieval, branded PDF generation, and a fixed-purpose host shutdown helper.
- Project overview, installation guide, quick reference, profile handbook, third-party component notice, screenshots, and reproducible build instructions.

Changed

- Delayed RetroPie receiver startup until EmulationStation finishes its initial device scan, preventing the virtual controller from disrupting cabinet input.
- Made the controller start disarmed and kept tracking/dashboard operation alive when RetroPie or the camera temporarily becomes unavailable.
- Moved private device settings and downloaded model data outside the App Lab installation ZIP.
- Changed the App Lab installation ZIP to download Google's Hand Landmarker model on first launch and verify its pinned SHA-256 digest before installation.

Fixed

- Recovered cleanly from USB camera disconnects and slow UVC camera wake-up.
- Kept the vision loop responsive through receiver, DNS, Wi-Fi, and mDNS loss.
- Corrected password pairing so credentials never appear in command arguments and the shared token is transferred and installed reliably.
- Corrected UNO Q dependency isolation, secure token upload, PDF builder file mode, landscape diagnostics layout, and cabinet launch integration.

Security

- Added short-lived TLS pairing with certificate comparison, a physical single-use PIN, bounded handshakes, and restricted token-file permissions.
- Required confirmation and a fixed host-side request path for system shutdown.
- Added a third-party component notice covering licenses, provenance, pinned versions, checksums, and update procedure.